THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. 125 × 8,000 = 1

(a) 1,00,000 **(b)** 10,08,000 **(c)** 10,00,000 **(d)** 1,00,00,000

2. If we add 1 to the largest 5-digit number we get: 1

(a) 99,999 **(b)** 10,00,000 **(c)** 1,000 **(d)** 1,00,000

3. 34,56,000 + 12,34,000 = 1

(a) 47,90,000 **(b)** 46,80,000 **(c)** 46,90,000 **(d)** 22,22,000

4. 17 + 25 + 3 is easily found by adding: 1

(a) 25 + 3 first **(b)** all three digit by digit

(c) 17 + 25 first **(d)** 17 + 3 first

5. Evaluate (6 + 10) × 5. 1

(a) 56 **(b)** 85 **(c)** 80 **(d)** 74

6. The value of the digit 8 in 7.806 is: 1

(a) 0.8 **(b)** 0.08 **(c)** 8 **(d)** 80

7. 5 cm 3 mm in centimetres is: 1

(a) 5.03 cm **(b)** 0.53 cm **(c)** 53 cm **(d)** 5.3 cm

8. The decimal 0.125 as a fraction is: 1

(a) 1/12 **(b)** 125/10 **(c)** 1/5 **(d)** 1/8

9. Simplify 6x + 3x − 1x. 1

(a) 8x **(b)** $8x^2$ **(c)** 17x **(d)** 10x

10. If a matchstick pattern uses 3n + 1 sticks for n squares, then 5 squares need: 1

(a) 18 **(b)** 15 **(c)** 16 **(d)** 21

11. The complement of 62° is: 1

(a) 118° **(b)** 152° **(c)** 28° **(d)** 62°

12. If a transversal makes alternate angles equal, the two lines are: 1

(a) equal **(b)** perpendicular

(c) parallel **(d)** intersecting at 90°

13. Angles around a point add up to: 1

(a) 90° **(b)** 720° **(c)** 360° **(d)** 180°

14. In the Virahāṅka sequence 1, 2, 3, 5, 8, … the term after 34 is: 1

(a) 56 **(b)** 54 **(c)** 55 **(d)** 68

15. The number 1729 is famous as the: 1

(a) Kaprekar number **(b)** smallest prime
(c) Hardy–Ramanujan number **(d)** square number

16. The line joining a vertex to the midpoint of the opposite side is the: 1

(a) diameter **(b)** bisector **(c)** altitude **(d)** median

17. A triangle with all sides different is called: 1

(a) isosceles **(b)** scalene **(c)** equilateral **(d)** right

18. The number of altitudes in a triangle is: 1

(a) 4 **(b)** 1 **(c)** 3 **(d)** 2

19. The reciprocal of 3/5 is: 1

(a) -3/5 **(b)** 3/5 **(c)** 5/3 **(d)** 2/5

20. 7/8 ÷ 3/4 = 1

(a) 6/7 **(b)** 1/8 **(c)** 21/32 **(d)** 7/6

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. The population of a town is 2,48,750. Round it to the nearest thousand. 2

22. Rewrite 36 + 24 as a multiple of a common factor. 2

23. Arrange in ascending order: 0.5, 0.05, 0.55, 0.505. 2

24. Identify the coefficient and constant in 8k − 5. 2

25. The angles around a point are 90°, 120°, 100° and x. Find x. 2

26. Why is 0 considered an even number? 2

27. State the condition for constructing a triangle with sides a, b, c. 2

28. Find 1/3 of 1/4. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. A shop sold goods worth ₹3,45,000, ₹2,78,500 and ₹4,06,250 in three months. Find the total. 3

30. Write four decimals between 0.1 and 0.2. 3

31. A bus carries 40 passengers. At each stop x get off and y get on. Write the number after one stop. 3

32. Two parallel lines are cut by a transversal. Alternate angle is (2x + 10)° and (x + 40)°. Find x. 3

33. Find x in a triangle with angles 2x, 3x and 4x. 3

34. A bus takes 1 1/2 hours for 63 km. Find its average speed. 3

35. A rectangle has length 24 m and width 15 m. Write an expression for its perimeter and evaluate. 3

OR

Estimate 49 × 51 using the expression (50 − 1)(50 + 1) pattern or by rounding. 3

36. Explain why any number and its digit-reverse differ by a multiple of 9 (two digits). 3

OR

Write the first 6 triangular numbers and their pattern. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) A rope is 15.5 m. Pieces of 1.25 m are cut. How many pieces? 5
(ii) What is left?
(iii) Express leftover in cm.

OR

Case study: A tank has 250.5 L. In the morning 48.75 L is used, in the evening 62.4 L. 5
(i) Total used.
(ii) Left.
(iii) Fraction used (approx).
(iv) Is it more than half?

38. (i) Find the angle between the hands at 4:00, 6:30. 5
(ii) At 3:30.
(iii) Explain the 6:30 case.

OR

(i) Six angles around a point are x, 2x, 3x, 3x, 4x and 5x. Find x. 5
(ii) Find the largest angle.
(iii) Is it obtuse?

39. (i) Show that the sum of exterior angles is 360°. 5
(ii) Find the third exterior angle if two are 130° and 120°.
(iii) Check.

OR

(i) The vertex angle of an isosceles triangle is 40°. Find each base angle. 5
(ii) Name the type of triangle.
(iii) Find an exterior angle at the vertex.

40. **Case study:** A tailor needs 1 3/4 m of cloth per shirt. He has 21 m. 5
(i) Shirts.
(ii) Cloth left.
(iii) If each shirt sells for ₹450, income.
(iv) Cost of cloth at ₹120/m.

OR

(i) Why does multiplying by a proper fraction reduce a number? 5
(ii) Why does dividing by a proper fraction increase it?
(iii) Give examples.

END OF QUESTION PAPER — Total: 80 Marks

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